

25th Euro Working Group on Transportation Meeting (EWGT 2023) Configurations and features of demand responsive transports

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Abstract

The concept of Demand Responsive Transport (DRT) has been around for more than 40 years and is a promising mobility alternative when traditional public transport proves inadequate in terms of its effectiveness or efficiency, as is the case of low-density areas. DRT systems have a wide range of operational configurations, being highly adaptable to different contexts and environments. Therefore, the design of a DRT mobility solution can become a quite complex and challenging problem. To assist in the design of DRTs, this paper aims to present a comprehensive classification of DRT features and to identify some common design choices in different operational scenarios. The proposed classification is based on a review of reports from available literature and previous European DRT projects. In addition, an analysis of the most usual configurations for different purposes and scenarios is presented. In this research, the operational, demand, and administrative characteristics of DRTs are addressed. Demand aspects encompass features that directly influence trip demand, such as service areas, target passengers, and hours of operation. Operational features include characteristics that will affect daily operations as the type of stops, frequency of the operation, booking methodology, vehicle route, pick-up and drop-off locations, and the vehicle type used. Administrative characteristics address the relationship between consumers and the system, such as the purpose of the system, fares, visual identification of stops, and booking methods. Regarding the usual design choices, our survey shows that rural DRTs are primarily oriented to serve populations in need in low-density areas and to complement existing PT gaps, while urban DRTs are mainly viewed as a mobility alternative to fill existing PT gaps. Defining design patterns for peri-urban and multi-area DRTs presents challenges due to their transitional nature, thus combining attributes of both rural and urban systems.

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1. Introduction

Traditional urban transportation systems are usually designed for high-density areas, in which the dense population allows for a great number of mobility alternatives. By contrast, in regions with a low population density, such as rural and peri-urban areas, the availability of transportation is highly impaired since traditional public transportation systems (PT) struggle to find an economically viable operation. PT systems usually reduce the number of stops and their frequency in these areas, lowering the costs to make the operation economically viable (Gomes *et al.*, 2015).

This approach heavily limits the access of the overall population to transportation and, therefore, their access not only to basic services (such as healthcare and educational centres) but also to opportunities for work and leisure activities (Brovarone & Cotella, 2020).

In this context, Demand Responsive Transport (DRT) is seen as a promising mobility alternative when traditional PT is not effective, especially in low-density areas. As Mageean and Nelson (2003) explain, DRT systems are a type of flexible transport and can be considered an intermediate form of transport, keeping the “on-demand” characteristics of the taxi while also preserving the “ridesharing” characteristics of the bus.

However, the heterogeneity of low-density areas turns the design and the evaluation of DRTs into a difficult and complex task, as Papanikolaou *et al.* (2017) point out. The design of a DRT mobility solution can, therefore, become quite a complex and challenging problem. To aid in the development of these systems, some studies propose methodologies to characterize transport systems, such as Ambrosino *et al.* (2000) that grouped these services in terms of increasing flexibility, from systems with predefined timetable and route services (like traditional PT), all the way to fully flexible systems (that can reach any origin/destination point chosen by the passenger). Takeuchi *et al.* (2003) propose grouping these systems according to the type of routes, timetable, and trip reservation methods.

Zografos *et al.* (2008) present a different perspective on the classification of DRT, analysing their features from a business model perspective.

These features are grouped into main characteristics such as business model dimension, context, service offering, business strategy, and service functions. Subsequent works, such as Denmark (2012) and INTERREG EUROPE (2018), have proposed similar methods for classification, categorizing DRT features based on aspects such as routing, stopping points, vehicles, bookings, passengers, fares, and operators. Moreover, Peterson (2019) has carefully examined real-world DRT systems, with some of these systems being analysed in this study. Peterson's work has focused on aspects such as virtual and physical stops and has proposed new classifications to define routing.

Winter *et al.* (2018) explore the capabilities of DRT systems operated by autonomous vehicles (ADRTS). Despite the complete automation of the vehicle fleet, this service retains numerous features typically associated with a DRT. The authors propose the following four service dimensions to classify the attributes of these systems: accessibility (stop-to-stop or door-to-door service); directness (predefined routes with transfer or stops or direct services); vehicle sharing (shared or sequential trips); demand responsiveness (restrictions of flexibility or passenger's waiting time).

Jittrapirom *et al.* (2019) conducted a comparison between DRT services in the area of Nijmegen, Netherlands. For this comparison, the authors classified the systems based on their ride-sharing aspects, routing approach, reservation methodology, access by the general public, wheelchair accessibility, pricing (fare cost), elderly discount, pick-up and drop-off locations, and waiting times.

Freiberg *et al.* (2021) present a detailed review of other methods, having developed a new classification proposal, that we have used as a basis for elaborating the classification presented in this work. Despite being very detailed, their classification is focused mainly on urban systems and lacks the categorization of some components of DRT design, such as pick-up and drop-off locations and operating hours. In a more recent study, Vansteenwegen *et al.* (2022) present a unifying classification framework for DRT as systems, based on the degree of their responses: dynamic online; dynamic offline; and static. Each category is related to a set of features present in these systems.

As depicted above, the literature presents some methodologies for classifying DRTs and their features. Although detailed, these classifications do not cover some key aspects of DRT design, such as the proximity of the pickup location to the passenger's position, and the main purposes of the system. Furthermore, these classification methodologies lack the ability to present how these design choices relate to the functioning of the system. Therefore, it is necessary to develop a general and comprehensive method for classifying DRT features, capable of reflecting the impact of these design choices in the operation of the transport.

This work presents three main contributions: First, a general and complete classification of DRT features is presented, divided into three main categories, according to their implications to the system. Second, an analysis of the most usual purposes of this system in three different service areas: rural, urban, and peri-urban/multi-area regions. Lastly, an analysis of the most usual DRT configurations in these three service areas.

The output of this work will integrate a decision support system (DSS) to assist in the planning of DRTs for low-density areas, supporting the design of these transportation systems from a tactical level, and suggesting the most appropriate configurations for the scenario presented and planned objectives.

2. Methodology

Aiming to understand the most usual configurations of DRTs in different scenarios, we started by describing and categorizing the main features of DRT systems. For this purpose, a sample of real-world DRT was put together, collected from reports of flexible transport projects such as SMARTA (The SMARTA Consortium, 2019), RESPONSE (Kirsimaa & Suik, 2020), SAMPO (Finn, 1996), and MAMBA (Dick *et al.*, 2020), in addition to the works by Buntinx *et al.* (2019), Freiberg *et al.* (2021), Haglund *et al.* (2019), and Enoch *et al.* (2019).

The analysed sample comprises a total of 52 DRT systems, operating in 24 different countries, as shown in Figure 1. Among these, 15 were designed for rural areas, 11 for urban areas, 6 for peri-urban areas, and 20 for operating in multiple areas. At least 30 of these DRTs were initially designed as pilot projects, with 17 successfully transitioning into regular operational services, while the remaining 13 ceased operation after the trial period.

Out of the 52 DRTs analysed, 35 were still in operation, while the remaining 17 had either concluded or suspended their operations (some DRTs suspended their operation due to the COVID-19 crisis, and subsequent re-establishment did not occur until March 2023). The average lifespan of DRT pilot projects that ceased their operation was about 18 months.

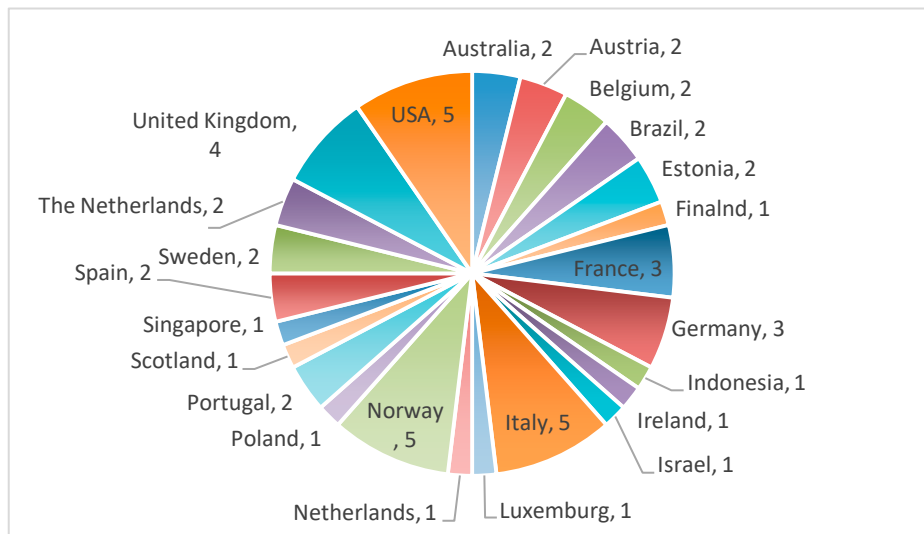


Fig. 1. Countries of the reviewed services

As we analysed the configuration of these reported DRTs, we were able to define three main groups of DRT features, namely:

- characteristics that affect the demand;
- characteristics that affect daily operation;
- administrative decisions.

Table 1 presents these three major sets of attributes and the characteristics that compose them. It is important to emphasize that the categories of each feature are not necessarily exclusive. Some DRTs combine some of these into

their operation, such as both types of stops or multiple service areas.

Table. 1. Proposed methodology for classifying DRT features

Main group	Feature	Categories
Characteristics that affect the demand	Service hours	Period of the day in which the service is available e.g., conventional transport hours, peak hours, off-peak hours, night services, and variable hours.
	Service area	The main area in which the service operates: urban, peri-urban, rural, or multi-area.
	Target passengers	The main group of passengers that the transport was designed to serve: all passengers or passengers in specific segments.
Characteristics that affect daily operation	Type of stops	A DRT can operate with fixed or variable stops.
	Frequency of the operation	Operation frequency. Some systems operate only when required, others have an established number of trips to make per day.
	Required booking lead time	Some systems establish a minimum period for the trip to be booked before the pick-up, while others allow the request on the spot.
	Vehicle route	Methodology and operational principles adopted for the vehicle route. Among the most common are established routes, established routes with diversions, and free-floating between stops.
	Pick-up and drop-off locations	Refers to the proximity of the actual pick-up and drop-off in comparison with the user's request: specifically requested point, close to the requested point, and existing mapped point.
	Vehicle type	Determines the capacity of simultaneous passengers a vehicle can operate. Common vehicles are minibusses, vans, buses, cars, and taxis.
Administrative decisions	Purposes of the service	The purposes of the DRTs design and implementation. Common purposes will be detailed in section 3.1.
	Operating methodology	System design methodology: one-to-one, many-to-one/one-to-many, or many-to-many.
	Fare methodology	How the fare is paid: a prior (by credit/debit cards or pre-paid tickets), when boarding, or free.
	Methodology for booking	The channels that passengers can use to book a trip. Usually, telephone and web (site/application) are available.
	Visual identification of stops	Some services use physical stops or visual identification to physical stops to assist passengers, they can be classified as physical stops or virtual stops.

The first set of attributes comprises factors that play a substantial influence on the number of trips that the system will need to handle during its operation. This category includes: the service hours; the service area; and the target passengers. Systems operating near or at peak hours will need to handle a larger volume of requests than those systems that operate at off-peak hours, weekends, and overnight periods. Moreover, densely populated areas, such as large urban centres, typically experience a higher number of trips, as opposed to less populated areas, such as rural

and peri-urban regions. Additionally, the system's target passenger group will also have an impact on the number of requests, according to the specific needs of the population segment under consideration (such as the most vulnerable people or those without access to existing regular transport services).

The second group encompasses features that directly affect the daily operation of the system. These include the type of stops, frequency of the operation, required booking lead time, the vehicle route, the pick-up and drop-off locations, and the vehicle type used in the operation. The type of the stop may be either fixed (like existing bus stops) or variable (such as street crossings or a place indicated by a phone operator/application). Another characteristic is related to pick-up and drop-off places, which may be an existing mapped point (physical or virtual), a point close to the request of the passenger, or the specific origin of the request (as in the case of door-to-door services). These final two attributes, in conjunction with the vehicle route flexibility (an established route, an established route with deviations, or free-floating between points), play a major role in the overall flexibility of the system.

The capacity of the vehicle type naturally bounds the number of simultaneous passengers it can serve. Minibuses, for instance, can attend to multiple passengers simultaneously, while smaller vehicles, such as taxis, may need to refuse a trip request due to their low passenger capacity. A final characteristic of this second group is related to the anticipation required for booking a trip. Imposing an advance notice for travel requests allows operators to optimize routes. However, the possibility of requesting a trip closer to the intended departure time is more attractive to the passenger, increasing the possibility of using the service.

The last set of attributes is composed of decisions of an administrative nature, which shape the structure of the service and influence the passengers' interaction with it. This category encompasses the operating methodology itself (one-to-one, many-to-one, or many-to-many), the purposes of the service, the visual identification of stops, how the fare is charged, and the methodology adopted for booking. The operating methodology and main purpose of the service will guide the choice of its features, while the ease with which passengers make reservations, pay for the trip, execute payments, and identify collection points, are crucial aspects for the adoption of the system by the population.

3. Analysis and Discussion

When analysing the configurations of the DRTs, it was possible to notice that those systems that operate in similar service areas usually have comparable configurations. Therefore, for this analysis, DRTs were put together into three groups, according to their service area: rural DRTs; urban DRTs; and peri-urban/multi-area DRTs. Since peri-urban areas are usually transitional areas between urban and rural regions, these were grouped together with DRTs that operate in more than one area.

3.1. Purposes of the service

As with other transport solutions, DRTs are developed and deployed with a set of purposes in mind. Identifying the problems and needs of a region is naturally the first step to designing an effective transportation system.

Through the analysis of the description reports of the 52 DRTs, we have identified and compiled a list of nine common primary purposes: (A) reduce operational costs; (B) to be a mobility alternative; (C) feed other transportation services; (D) reduce pollution; (E) serve populations in need; (F) complement existing PT gaps; (G) provide transport in rural areas; (H) connect rural/peri-urban/urban areas; and (I) provide touristic services. It is important to acknowledge that the design methodology of a DRT can be multi-purpose, addressing different aspects of its implementation. The distribution of frequency for these purposes is presented in Figure 2.

The most common purposes of DRTs operating in rural areas are to complement existing PT gaps, to provide transport in rural areas, and to serve populations in need. For those in urban areas, the main purposes found were the creation of alternative mobility and complementing existing PT gaps. For DRTs operating in transition areas, the more common purposes included being a mobility alternative and complementing existing PT gaps, in addition to the use of these DRTs as feeders, allowing the population access to other transport systems.

For the analysed DRTs in rural areas, the most common purposes are in line with the most common transportation problems reported in the literature in these regions, such as the lack of traditional PT or inefficiencies due to the challenges caused by low-population density. In urban areas, the use of DRTs is usually seen as a complement to the existing PT network, commonly outside peak hours, or as an alternative to the use of private vehicles. In transition

areas, there is a combination of the mentioned purposes, plus an extensive use of these systems as feeders of other transport networks.

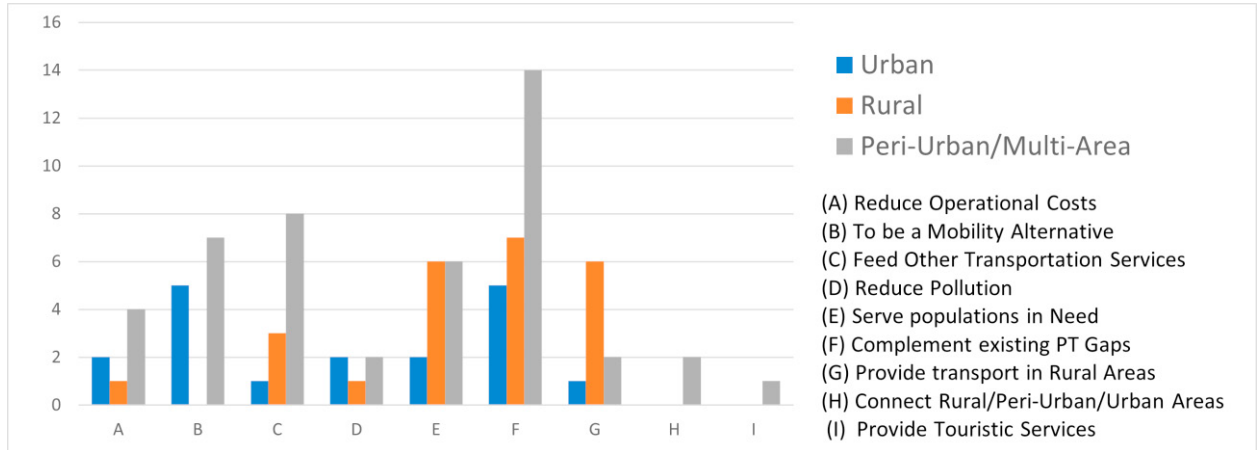


Fig. 2. Main DRT purposes.

3.2. Type of stops and pick-up locations, frequency of operation, vehicle route and time, and booking methodology.

Services that operate in rural areas are commonly designed to be completely or partially door-to-door, picking up and dropping off passengers at their exact intended origin and destination. Due to this characteristic, variable stops are usually, virtually placed in the requested places. Free-floating between stops is also the most common approach for designing vehicle paths. Moreover, most of the services require booking trips in advance, with advance notice varying from 1 to 24 hours. Booking via a call centre is present in all systems except one, while dedicated apps are available in about half of them. Most of the analysed services operated on regular transport hours and only when requested.

Regarding DRTs that operate in urban areas, the most common design is the so-called “corner-to-corner”, or some variation of this method; the passenger is directed to a nearby pick-up point (within walking distance), which can either be a bus stop of the regular PT service (with or without identification of the DRT service), a given avenue, or an intersection of streets. Therefore, there is no prevalence between physical or virtual stops in the analysed DRTs. Vehicles mostly operate free-floating routes between stops. Concerning the booking procedure, about half of the services allow at-the-stop travel requests, while the other half requires advanced reservations (varying between 30 minutes and 48 hours before traveling). In general, many services include vehicles adapted for wheelchairs. These services mostly operate during PT's conventional hours (except one, which operates only during the early hours of the morning).

As for the peri-urban areas and DRTs operating in more than one area, there is, in general, a mix of characteristics, thus making it difficult to define some kind of standard. Most of these systems share some features, including a design thought for the use of the general population and the implementation of free-floating routing.

Among the DRTs whose main purpose is to complement the existing PT, it is usual to apply "corner-to-corner" schemes, with the use of existing bus stops for the collection of passengers. These characteristics are also somehow present in DRTs that aim to be an alternative mobility to the existing transport. Finally, DRTs to serve the population in need replicate some of the characteristics of rural systems and are designed in general as a "door-to-door" system with virtual stops.

3.3. Target passengers

About 40% of the rural DRTs analysed are targeted to a specific niche population, although many allow the general public to ride them. As referred in Section 3.1, serving a population in need is one of the most common purposes of

DRT services in rural areas. Usually, these populations are composed of elderly, impaired people, or people in vulnerable social conditions. The requirements of this target population are reflected in the design choices, as in the door-to-door service and the large presence of call centre booking options.

Regarding DRTs operating within urban areas, out of the 11 analysed systems, only two are designed with the purpose of serving a specific populational group. As established earlier, the prevalent purposes shared by urban DRTs are to be a mobility alternative and to complement the existing transportation gap. Consequently, these systems also serve as an alternative to private vehicles, offering passengers faster and more comfortable trips, when compared to traditional PT. To make the operation viable, the route is optimized through small changes in the pick-up and drop-off points requested by passengers. Moreover, urban (and some peri-urban DRTs) commonly deny booking trips that can be done using the existing PT infrastructure.

3.4. Fares and vehicles

Among the analysed systems, four different fare procedures have been adopted: free; fixed-costed fares; fares based on the distance of the ride; and fares based on the pick-up time. By far, the most adopted method is fixed fares, present in 38 of the 52 DRTs. Moreover, some DRTs had more than one fare category, depending on the age or socio-economic status of the passenger. Fares calculated based on travel distance accounted for 10 of the 52 DRTs, while those considering pick-up time were only 3 of the total. However, some DRTs that applied these last two procedures still had fixed tariffs in certain situations (such as off-peak hours and pick-ups at bus and train stations). The only 2 cases of free tariffs were in rural DRTs.

Although minibuses and vans are the most used vehicles in DRT systems, there was no clear pattern of usage in this set. Other characteristics besides those studied probably influence the choice of vehicle to be used in the operation, such as the availability of vehicles in the existing fleet, the condition and configuration of the traffic area, and the budget available for the acquisition of vehicles.

4. Final Remarks

In this work, we have analysed a heterogeneous group of 52 DRTs, aiming to establish a detailed and complete classification of DRT features. Furthermore, we were able to apply the proposed classification methodology to determine the most usual and efficient designs for DRTs in different operating scenarios. Through a detailed examination of the sample, we were able to organise their multiple features into three main classification groups: characteristics that affect the demand; characteristics that affect the daily operation; and administrative decisions.

Afterward, when analysing the sample of DRTs, we noticed that those operating in similar service areas had comparable configurations. Therefore, to analyse the most common configurations, we divided DRTs into three groups: rural; urban; and peri-urban/multi-area.

By categorizing these DRT systems into distinct groups, we managed to recognize that they share common purposes. We have also carefully analysed the most common purposes of these systems. Subsequently, the features of each DRT are tailored to accommodate these purposes. For instance, rural DRTs typically aim to assist populations in need (elderly, children, impaired, or low-income people) who are often capable of planning their journeys well in advance. Consequently, these systems prioritize door-to-door transportation services with anticipated booking (usually via telephone). On the other hand, one of the urban DRT's primary purposes is to be a mobility alternative to private cars. Therefore, these systems are more oriented on providing at-the-spot booking, with wider use of technology, while providing corner-to-corner services.

It should be noted that, in this analysis we did not differentiate between fully developed services and pilot projects. These last systems are usually developed in the context of an experimental study, with guaranteed funding throughout the study duration. Moreover, there is a need for a more in-depth analysis of the characteristics of DRTs, not only during the pilot project phase but also when they become regular services.

In addition, there are some promising avenues for future research, such as analyse the operational characteristics of these systems and user satisfaction in different scenarios and purposes. In the literature, the authors agree that the demand for these systems is highly correlated to their level of service. Combined with the outcomes of this research,

this knowledge can be integrated into a decision support system for the design of more effective, socially inclusive DRT services.

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